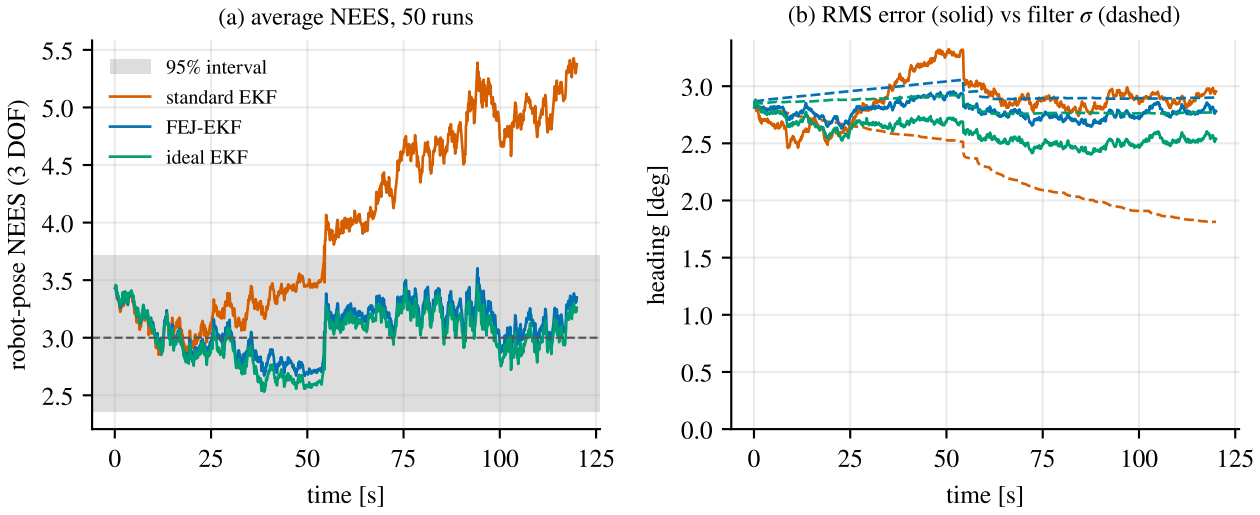

(a) average NEES, 50 runs
95% interval
standard EKF
FEJ-EKF
ideal EKF
robot-pose NEES (3 DOF)
5.5
5.0
4.5
4.0
3.5
3.0
2.5
0
25
50
75
100
125
time [s]
(b) RMS error (solid) vs filter σ (dashed)
heading [deg]
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
25
50
75
100
125
time [s]